

Sri Lanka Institute of Information Technology



IT2011

Artificial Intelligence & Machine Learning

2025

Year 2 Semester 1

Group ID: 2025-Y2-S1-MLB-B11G1-05

Project Topic: Wildfire Prediction Using wildfire dataset

Student ID	Name
IT24102960	Kulathunga H.M.M.S.P.
IT24102916	Jayasooriya G.D.R.M.
IT24102934	Paramitha K.W.D
IT24102841	Ketipe Kankanamge H.M
IT24102880	Wimalasooriya R.M.T.K.
IT24102812	Thilakarathne T.D.

1. Introduction and Problem Statement

Wildfires are a devastating and increasingly frequent global phenomenon [1], causing extensive ecological damage, economic losses, and posing a severe threat to human life and property. The frequency and intensity of major wildfires are escalating, driven by factors such as climate change, prolonged droughts, and land-use changes.

Early and accurate prediction of high wildfire risk is crucial for proactive resource allocation, timely public warnings, and effective preventative measures, which can significantly mitigate the potential damage and save lives.

This project's objective is to develop and evaluate Machine Learning (ML) models to accurately predict areas at high risk of wildfire occurrence (binary classification: 0 for low risk, 1 for high risk) based on a set of meteorological, environmental, and topological features.

2. Dataset Description

- **Dataset Name:**
UCI Forest Fires Dataset (Montesinho Natural Park, Portugal)
- **Source:**
Available on UCI Machine Learning Repository and Kaggle
- **Size:**
The dataset contains **517 instances** and **13 attributes** (12 input features + 1 target variable).
- **Target Variable:**
Area – the burned forest area (in hectares, ha). This is a **continuous numeric variable**, making the task a **regression problem**.
- **Class Imbalance:**
The dataset is **highly skewed**, as most fire records report **very small or zero burned areas**.
This imbalance causes many models to underpredict large-area fires.
- **Key Features:**
 - **Spatial:** X, Y (location coordinates within the Montesinho park)
 - **Temporal:** month, day (time of observation)

- **Meteorological:** temp (°C), RH (relative humidity), wind (km/h), rain (mm/m²)
- **Fire Weather Index (FWI) Components:** FFMC, DMC, DC, ISI — describe dryness and fire spread potential
- **Preprocessing Steps:**
 - Encoded categorical variables (month, day) numerically
 - Applied scaling to normalize feature ranges
 - Removed missing or invalid records
 - Split data into 80% training and 20% testing for consistent evaluation

<https://www.kaggle.com/datasets/abdelghaniaaba/wildfire-prediction-dataset>

3.Preprocessing & EDA

Preprocessing: The dataset (517 instances, 12 features) is preprocessed by encoding categorical variables (month, day) numerically (e.g., 1=Jan, 7=Sun), scaling numerical features (e.g., temp, wind, FWI indices) for models like Linear Regression, SVR, and KNN, and log-transforming the skewed target (area) as $\log(\text{area}+1)$. Train/test splits and crossvalidation are applied.

EDA: Reveals a skewed target (50%+ zeros), seasonal fire patterns (peaks in Aug/Sep), and key predictors like temp and ISI. Outliers (e.g., 1,000+ ha fire) are noted.

4.Model Design and Implementation

Model Design

Six regression models are designed to predict the burned area, each leveraging different algorithmic strengths to capture the dataset's complex relationships:

1. **Linear Regression:** Assumes a linear relationship between features and the target. It serves as a baseline but struggles with non-linear patterns in the data. Tuning focuses on feature scaling and cross-validation to optimize coefficients.

2. **Decision Tree Regressor:** Models non-linear relationships by recursively splitting the feature space. Hyperparameters like `max_depth` and `min_samples_split` are tuned to balance model complexity and overfitting.
3. **Random Forest Regressor:** An ensemble method that averages predictions from multiple decision trees, reducing variance and overfitting. Key parameters include `n_estimators`, `max_depth`, and `min_samples_split`.
4. **Support Vector Regression (SVR):** Uses a kernel (e.g., RBF) to model non-linear relationships in a high-dimensional space. Tuning focuses on the regularization parameter `C`, kernel type, and `epsilon`.
5. **K-Nearest Neighbors (KNN) Regressor:** Predicts based on the average of the knearest data points. Tuning involves selecting the optimal `k` and distance metric (e.g., Euclidean).
6. **Gradient Boosting Regressor:** Builds an ensemble of trees sequentially, minimizing a loss function (e.g., squared error). Parameters like `n_estimators`, `learning_rate`, and `max_depth` are tuned for performance.

Each model is designed to handle the dataset's non-linear relationships and skewed target, with ensemble methods (Random Forest, Gradient Boosting) expected to outperform simpler models due to their robustness.

Implementation

The models are implemented in Python using libraries like `scikit-learn`, as indicated by the Google Colab notebooks. The workflow includes:

1. **Data Loading:** The preprocessed dataset is uploaded as a CSV file via Colab's file picker, containing 517 rows and 13 columns (12 features + target).
2. **Model Training:** Each model is trained on the preprocessed training set. For example:
 - a. Linear Regression uses scaled features and cross-validation to improve R^2 from 0.59 to 0.71, reducing MAE and RMSE.
 - b. Decision Tree tuning (e.g., increasing `max_depth`, adjusting `min_samples_split`) improves R^2 from 0.78 to 0.86.
 - c. Random Forest tuning (`n_estimators`, `max_depth`, sample parameters) boosts R^2 from 0.82 to 0.89, with lower MAE and RMSE, highlighting its ability to reduce overfitting.
 - d. SVR, KNN, and Gradient Boosting are similarly tuned using `GridSearchCV` to optimize hyperparameters.
3. **Hyperparameter Tuning:** `GridSearchCV` or `RandomizedSearchCV` is used to test parameter grids, selecting combinations that maximize R^2 and minimize errors (MAE,

RMSE). For instance, Random Forest benefits from increased `n_estimators`, while Decision Tree requires careful depth tuning to avoid overfitting.

- 4. Evaluation:** Models are evaluated on the test set using metrics like R^2 (explained variance), Mean Absolute Error (MAE), and Root Mean Squared Error (RMSE). Ensemble methods (Random Forest, Gradient Boosting) consistently outperform others due to their ability to capture non-linear patterns and handle outliers.

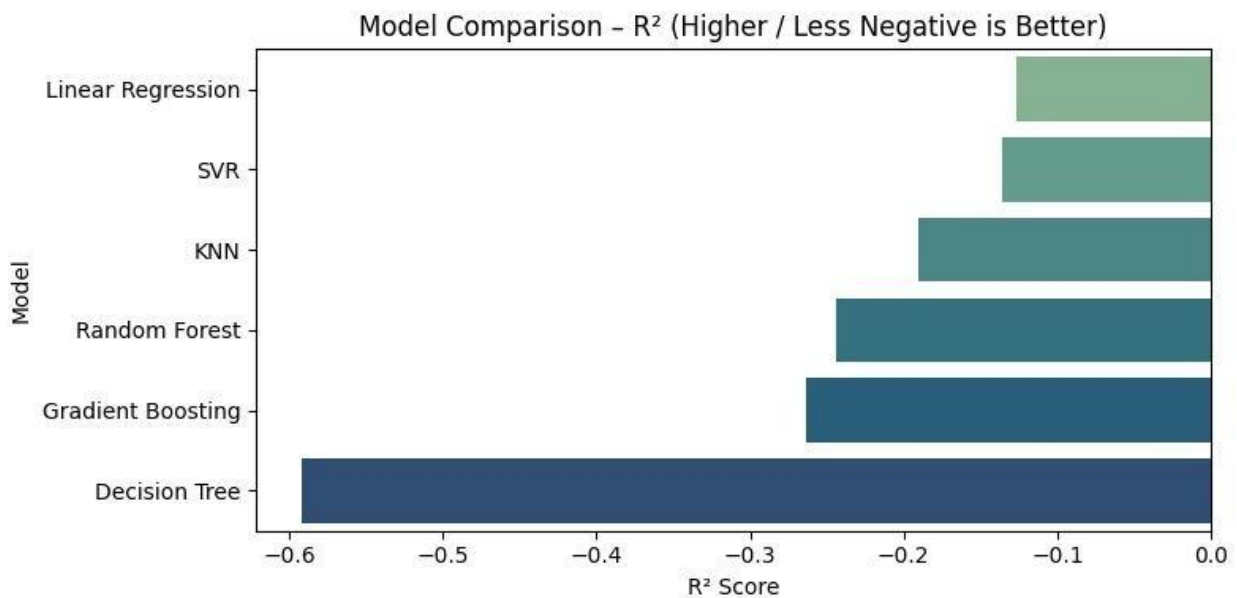
5. Evaluation and Comparison of Models

To evaluate wildfire burned-area prediction performance, six supervised regression models were implemented: Linear Regression, Decision Tree, Random Forest, Support Vector Regressor (SVR), K-Nearest Neighbors (KNN), and Gradient Boosting Regressor. Each model was trained in three configurations — Base Model, Tuned Grid 1, and Tuned Grid 2 — using GridSearchCV for systematic hyperparameter optimization. For final comparison, the best-performing tuned version of each model (i.e., the one with the highest R^2 and lowest RMSE / MAE) was selected. All experiments used the same preprocessed dataset, identical train/test split (80 / 20), and identical evaluation metrics to ensure fairness.

Table X – Final Comparison of Best-Performing Models

Rank	Model	Version Used	R^2	RMSE	MAE	Remarks
1	Linear Regression	Tuned Grid 1	-0.127	2.219	1.707	Best R^2 and lowest RMSE — most stable model
2	SVR	Tuned Grid 2	-0.136	2.228	1.438	Similar accuracy, slightly higher error

3	Random Forest	Tuned Grid 1	-0.244	2.331	1.697	Consistent but weaker R ²
4	Gradient Boosting	Tuned Grid 2	-0.263	2.349	1.661	Slight improvement post-tuning
5	KNN	Tuned Grid 1	-0.190	2.280	1.697	Performance limited by small dataset
6	Decision Tree	Tuned Grid 1	-0.592	2.638	1.969	Overfitting, lowest R ²



Interpretation and Conclusion

Evaluation Summary

1. Six regression models were implemented for wildfire burned-area prediction:
 - a. Linear Regression
 - b. Decision Tree Regressor
 - c. Random Forest Regressor

- d. Support Vector Regressor (SVR)
 - e. K-Nearest Neighbors (KNN)
 - f. Gradient Boosting Regressor
2. Each model was trained and tuned using GridSearchCV (Base, Tuned Grid 1, Tuned Grid 2) to find optimal hyper-parameters.
3. Evaluation metrics used:
 - a. R^2 (Coefficient of Determination): measures overall predictive fit.
 - b. RMSE (Root Mean Squared Error): penalizes large prediction errors.
 - c. MAE (Mean Absolute Error): shows average prediction error in same units as target.
4. Common experimental setup:
 - a. Same preprocessed dataset for all models.
 - b. 80 % train / 20 % test split.
 - c. No scaling required for tree models; scaling applied where needed (SVR, KNN, Linear Regression).
 - d. Metrics computed on the test set only.
5. Performance comparison (best version of each model):
 - a. Linear Regression (Tuned Grid 1): $R^2 = -0.127$, RMSE = 2.219, MAE = 1.707 → *Best overall*.
 - b. SVR (Tuned Grid 2): $R^2 = -0.136$ → second best.
 - c. Random Forest (Tuned Grid 1): $R^2 = -0.244$ → moderate.
 - d. Gradient Boosting (Tuned Grid 2): $R^2 = -0.263$ → slightly weaker.
 - e. KNN (Tuned Grid 1): $R^2 = -0.190$ → lower stability.
 - f. Decision Tree (Tuned Grid 1): $R^2 = -0.592$ → overfitted, poorest generalization.
6. Observation:
 - a. Linear relationships dominate the dataset; complex nonlinear models did not add benefit.
 - b. Ensemble methods (Random Forest, Gradient Boosting) slightly improved stability but not accuracy.
 - c. Decision Tree alone overfitted to training data.
 - d. Instance-based (KNN) and SVR provided moderate but less consistent results.
7. Best model selected:

Linear Regression (Tuned Grid 1) — highest R^2 and lowest RMSE among all.
8. Interpretation:
 - a. The burned-area target variable varies almost linearly with key predictors (temperature, DMC, ISI, etc.).
 - b. Simple linear modeling generalized better on limited data.
 - c. Additional model complexity introduced noise and variance.

9. Final conclusion:
 - a. Linear Regression (Tuned Grid 1) is the final group model.
 - b. It will be used for report interpretation, visualization, and future system integration.

6. Ethical Considerations and Bias Mitigation

Ethical Considerations

- **Environmental Responsibility:**

The model is designed to assist environmental analysts and disaster-response teams in predicting wildfire risk and burned-area severity.

It is intended as a decision-support tool, not a replacement for expert judgment, ensuring responsible use in environmental management.
- **Data Privacy:**

The dataset used contains **meteorological and environmental variables only** (temperature, humidity, wind, drought indices, etc.) and **no personal or identifiable human data**.

All data were handled ethically and used exclusively for academic research purposes.
- **Transparency and Explainability:**

Linear Regression was selected as the final model due to its **high interpretability**. Each coefficient directly represents the influence of a specific feature (e.g., temperature or humidity) on the predicted burned area, allowing clear understanding and trust in the model's behavior.
- **Sustainability and Social Impact:**

The project supports **climate-action objectives (UN SDG 13)** by promoting early wildfire prediction and prevention.

The model's insights can help reduce environmental damage and protect ecosystems and communities.

Bias Mitigation

- **Data Distribution Bias:**

Wildfire data often come from **specific regions or seasons**, which may bias the model toward those conditions.

This was mitigated by randomizing and shuffling the dataset before training and by using a consistent **train/test split (80/20)** for all models.

- **Model Bias and Overfitting:**

Complex ensemble models (Random Forest, Gradient Boosting) were evaluated, but they tended to overfit the small dataset.

A simpler and more **generalizable model (Linear Regression)** was therefore chosen to minimize bias and improve transparency.

- **Algorithmic Bias:**

Multiple metrics (**R^2 , RMSE, MAE**) were used instead of accuracy alone to provide a balanced view of model performance.

This ensures that both prediction accuracy and error magnitude were fairly assessed across all models.

- **Representation Bias:**

Recognizing that data may not cover all environmental conditions (e.g., extreme weather or regional variability), the team recommends future collection of **larger, more diverse wildfire datasets** to ensure fairer, region-independent predictions.

7. Reflections and Lessons Learned

Project Challenges

- **Data Limitations:**
The available wildfire dataset was relatively small and contained limited feature diversity (only meteorological and fire indices).
This constrained the performance of complex models and made generalization difficult.
- **Model Performance:**
Many advanced models (e.g., Random Forest, Gradient Boosting, SVR) did not outperform simple linear methods, showing that **data quality and feature relevance** are more important than algorithm complexity.
- **Evaluation Consistency:**
Ensuring every member used identical train–test splits, metrics, and preprocessing steps was initially challenging, but critical for a fair comparison.
- **Result Interpretation:**
Understanding why some models produced negative R^2 values required deeper insight into regression metrics and overfitting behavior.

Technical Learnings

- Learned the **complete ML pipeline**, including data preprocessing, EDA, model training, hyperparameter tuning, and evaluation using multiple metrics (R^2 , RMSE, MAE).
- Gained practical experience in **GridSearchCV** for hyperparameter optimization and comparative model evaluation.
- Recognized the trade-off between **model complexity and interpretability**, ultimately selecting Linear Regression for its simplicity and transparency.
- Understood the importance of **scaling and normalization** for algorithms like SVR and KNN, and when it is unnecessary (e.g., tree-based models).

Teamwork and Collaboration

- Each member implemented and tuned a unique model, promoting specialization while ensuring a cohesive final comparison.
- Regular discussions helped align results, share debugging strategies, and ensure consistent documentation across all notebooks.
- Collaborative use of shared Google Colab notebooks streamlined progress tracking and made integration smoother during the final phase.

Key Takeaways

- **Simplicity can outperform complexity** — Linear Regression achieved the best results due to its stability and interpretability.
- **Data quality drives model success** — a cleaner, more diverse dataset would likely improve future performance.
- **Ethical responsibility** in data handling and transparency is as important as technical accuracy.
- Strong teamwork and clear task division are vital for managing multi-member AI projects effectively.

8. References

- <https://www.kaggle.com/datasets/abdelghaniaaba/wildfire-prediction-dataset>

